

Deletion-Based Faithfulness Metrics Do Not Discriminate Between Attribution Methods on Volumetric CT

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Abstract—Attribution maps assert that the voxels they rank highest are those a model responded to. Deletion metrics test this by masking top-ranked voxels and recording the change in predicted probability. We apply the protocol to CT-FM, a SegResNet pretrained by contrastive self-supervised learning on 148,000 CT scans, fine-tuned for lung-nodule malignancy classification on volumetric chest CT, and compare three layer-wise relevance propagation composites against Grad-CAM. Two results are reported. First, over three paired seeds CT-FM initialisation gives a positive mean AUC gain over an architecturally identical randomly-initialised control at every label budget, though the gain is smaller than the between-seed spread at 10% and 100% of labels and the mean gain at full supervision is +0.006. Second, none of the four attribution methods achieved a lower deletion AUC than a random-order baseline, scoring 0.12 to 0.22 lower on AOPC. We attribute this to the masking scheme: random deletion distributes replaced voxels uniformly through the volume and moves the input away from the training distribution, whereas attribution-guided deletion removes a compact contiguous region. Under mean-value masking the baseline therefore benefits from distribution shift. Perturbation stability does separate the methods, with Grad-CAM at 0.924 and the LRP composites between -0.015 and 0.220 . A four-member deep ensemble reduces expected calibration error from 0.127 to 0.051 relative to MC dropout and attributes 53% of predictive entropy to model disagreement against 2.4% for dropout. We conclude that deletion protocols require manifold-preserving masking before they can rank attribution methods on volumetric medical data.

Index Terms—explainable AI, layer-wise relevance propagation, foundation models, self-supervised learning, uncertainty estimation, computed tomography

I. INTRODUCTION

Two claims are common in the medical imaging literature. The first is that self-supervised pretraining on large unlabelled image collections improves downstream performance. The second is that a published attribution map describes what the resulting model responded to. This paper evaluates both on a single task.

We fine-tune CT-FM [1], a SegResNet pretrained by contrastive self-supervised learning on 148,000 CT scans, for benign-versus-malignant lung-nodule classification on volumetric chest CT. We then explain it with layer-wise relevance propagation (LRP) [2] implemented in zennit [3], whose layer-type registry covers `Conv3d` and the 3D pooling operators and therefore applies to volumetric networks without modification. Grad-CAM [4] is included as a baseline.

Our contributions are:

- 1) A label-efficiency ablation against an architecturally identical from-scratch control at 10%, 25% and 100% of labels over three paired seeds, reporting the between-seed spread alongside the mean.
- 2) A faithfulness evaluation of three LRP composites and Grad-CAM on 3D CT against a random-order deletion baseline, with an analysis of the resulting negative outcome.
- 3) A measurement of explanation stability under input perturbation, and of the correlation between attribution faithfulness and predictive uncertainty.

II. METHOD

A. Data

NoduleMNIST3D [8] comprises 3D chest CT patches centred on lung nodules drawn from LIDC-IDRI [9] and labelled benign or malignant. Splits are 1,158 training, 165 validation and 310 test volumes, with a 863:295 benign-to-malignant training imbalance. We therefore use class-weighted cross-entropy, report balanced accuracy and AUC alongside accuracy, and select models on validation AUC.

We use the 64³ release. This is not cosmetic: CT-FM downsamples by a factor of 16, so a 28³ volume is not divisible by the network stride and raises an error.

B. Model and implementation notes

CT-FM’s encoder produces a five-level feature pyramid; we take the bottleneck at 512 channels and 4³ resolution, apply global average pooling, LayerNorm and a linear head. Three implementation details are recorded because each affects correctness without producing an error message.

First, the published checkpoint does not load into MONAI’s SegResNetDS. Parameter names differ, and `load_state_dict(strict=False)` matches *zero of 161* tensors without raising, so training proceeds on a randomly initialised network while the code appears correct.

Second, the from-scratch control must be architecturally identical. Constructing a second SegResNet from an inferred configuration yielded 19.9M parameters against the true 87.2M; that ablation would have measured capacity and

TABLE I
LABEL-EFFICIENCY ABLATION OVER THREE PAIRED SEEDS

Labels	CT-FM	Scr.	Mean gain	Gain range	Wins
10%	0.809	0.792	+0.016	$[-0.001, +0.039]$	2/3
25%	0.859	0.846	+0.014	$[+0.002, +0.035]$	3/3
100%	0.896	0.890	+0.006	$[-0.006, +0.022]$	2/3

attributed the difference to pretraining. Our control is the loaded model with every parameter reinitialised.

Third, pooling the full SegResNet output gives the linear head two features, since that output is a 2-channel volume. With unnormalised activations, whose standard deviation is approximately 0.03, training loss remained at 6.37 with AUC 0.50.

The encoder is fine-tuned at 10^{-5} and the head at 10^{-3} . Fine-tuning 87M pretrained parameters at the head’s learning rate displaces the pretrained representation, which is the quantity under evaluation.

C. Attribution

LRP redistributes the output score backwards under conservation rules and produces a map at input resolution. We compare three composites: ε -plus-flat, ε - γ -box with box bounds matching the $[-1, 1]$ input range, and ε - $\alpha 2\beta 1$. All three are reported, since the choice of rule affects the resulting map. Grad-CAM is computed on the encoder bottleneck and trilinearly upsampled.

D. Faithfulness, stability and uncertainty

Deletion progressively replaces the highest-ranked voxels with the dataset mean and tracks the target-class probability; a faithful map produces a steep early drop and hence a low area under the curve. **Insertion** restores highest-ranked voxels onto a blurred volume. **AOPC** is the mean probability drop across deletion steps. Every map is accompanied by a random-order baseline, because absolute values depend on model and data and only the gap against random is interpretable.

Stability is the mean correlation between the map on a clean input and on a noisy copy ($\sigma = 0.05$), noise the model itself ignores.

Uncertainty is estimated two ways. MC dropout keeps dropout active at inference and averages 20 forward passes. A deep ensemble averages four separately trained models, taken from the seed replicates of Section III-A at no additional training cost. Both give predictive entropy and mutual information, the latter being the share of entropy attributable to disagreement between samples or members, together with expected calibration error over 10 equal-width bins.

Sections IV-B and IV-F are computed on the 64 volumes of the attribution analysis; Section IV-E uses the full 310-volume test split, so its MC dropout figures differ.

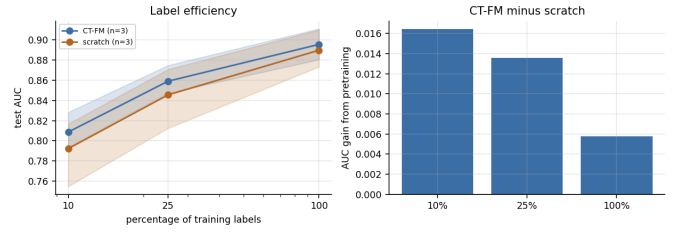


Fig. 1. Label efficiency and the mean AUC gain from pretraining. Shaded bands span the minimum and maximum over three seeds.

TABLE II
FAITHFULNESS AND STABILITY, 64 TEST VOLUMES

Method	Del.↓	Ins.↑	AOPC—rnd	Stab.
Random baseline	0.587	—	—	—
LRP ε +flat	0.819	0.972	−0.221	0.012
LRP $\varepsilon\gamma$ box	0.822	0.972	−0.224	−0.015
LRP $\alpha 2\beta 1$	0.816	0.973	−0.219	0.220
Grad-CAM	0.712	0.974	−0.119	0.924

III. RESULTS

A. Label-efficiency ablation

Three paired seeds were run at each label budget, with both arms of a seed using identical data subsets. Table I and Fig. 1 report the results.

The mean gain from pretraining is positive at all three budgets. At 10% and 100% of labels it is smaller than the spread between seeds, and one seed at each of those budgets favoured the randomly-initialised arm. Only the 25% budget places CT-FM ahead in every seed, with a mean gain of +0.0136.

The largest single-seed gain at full supervision, +0.0218, is approximately four times the three-seed mean of +0.0058. A single run of this experiment can therefore misstate the effect by a wide margin, which is relevant when comparable ablations are reported from one seed.

With three replicates these data support a small positive effect of self-supervised pretraining on this task and do not resolve its magnitude. Separating +0.006 from zero at full supervision would require substantially more replicates.

B. Calibration

Expected calibration error under MC dropout is 0.145 with mean predictive entropy 0.085 on the 64 volumes of the attribution analysis. Table III reports 0.127 for the same estimator on the full test split: the model is confident and less accurate than its confidence implies, the usual direction for a small fine-tuned network on an imbalanced task.

C. Faithfulness against a random baseline

A faithful map should sit *below* the random baseline on deletion AUC. All four sit well above it (Table II, Fig. 2): removing the voxels these methods rank highest damages the prediction *less* than removing random voxels.

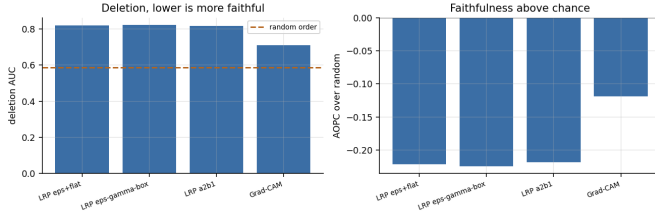


Fig. 2. Deletion AUC against the random-order baseline, and AOPC relative to random. All four attribution methods fall on the wrong side of the baseline.

TABLE III
UNCERTAINTY ESTIMATORS ON THE TEST SPLIT

Estimator	AUC	ECE	Entropy	MI share
Best single member	0.9111	—	—	—
MC dropout, 20 passes	0.8960	0.1265	0.1254	0.024
Deep ensemble, 4 members	0.9085	0.0505	0.3256	0.530

The likely cause is the masking scheme. Random deletion distributes replaced voxels uniformly through the volume; at 5% this produces a volume unlike any CT scan, and the network’s output falls because the input is far from the training distribution. Attribution-guided deletion removes a compact contiguous region, which remains closer to a plausible scan. Under this protocol the random baseline benefits from distribution shift.

Two consequences follow. Deletion with mean-value masking does not discriminate between attribution methods on this task, and a manifold-preserving scheme such as inpainting the removed region is required before the protocol can be used for ranking. Insertion also fails to separate the methods, with all four recovering to 0.972–0.974.

D. Perturbation stability

Under perturbation the methods diverge. Grad-CAM maps are nearly invariant (0.924) while the LRP composites give 0.012, -0.015 and 0.220 . Grad-CAM is computed at 4^3 and upsampled, which constrains how much the map can change. LRP resolves to input resolution and is correspondingly more sensitive to input perturbation. Neither metric reported here establishes which behaviour is preferable.

E. Deep ensemble and MC dropout

Four separately trained models were evaluated on the full 310-volume test split, with individual test AUC 0.8951, 0.9093, 0.9111 and 0.8805.

The ensemble reduces expected calibration error from 0.1265 to 0.0505 at comparable AUC (Table III, Fig. 4). The two estimators differ in how they apportion the uncertainty. Under MC dropout, mutual information accounts for 2.4% of predictive entropy, assigning almost all reported uncertainty to ambiguity in the data. Under the ensemble it accounts for 53.0%, indicating disagreement between independently trained models that dropout sampling on a single parameter set does not expose.

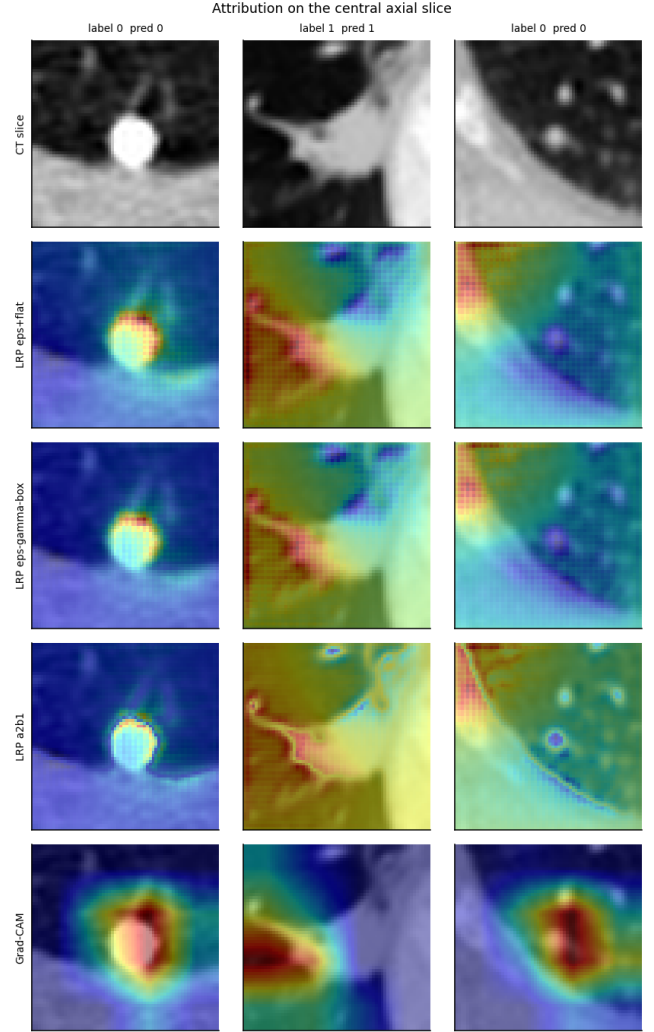


Fig. 3. Attribution on the central axial slice for three test volumes.

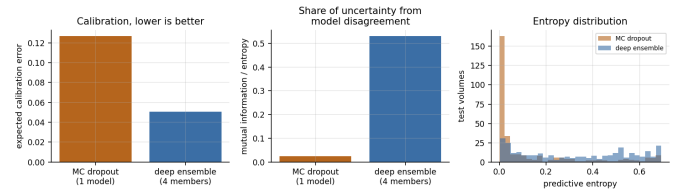


Fig. 4. Calibration error, the share of predictive entropy attributable to model disagreement, and the entropy distribution, for MC dropout against a four-member deep ensemble.

F. Uncertainty and attribution faithfulness

Spearman correlations between predictive entropy and per-sample AOPC are $+0.190$, $+0.174$ and $+0.221$ for the three LRP composites and -0.124 for Grad-CAM. The three LRP composites agree with each other and give weakly positive correlations; Grad-CAM gives a weakly negative one. With 64 volumes none of these correlations is statistically significant. The same question is posed for segmentation by Salahuddin

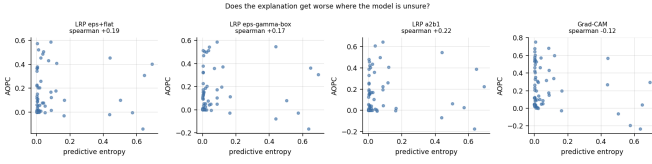


Fig. 5. Per-sample predictive entropy against AOPC. Correlations are weak and the three LRP composites agree with one another and disagree with Grad-CAM.

et al. [6].

IV. LIMITATIONS

The ablation uses three paired seeds. The mean gain is smaller than the between-seed spread at 10% and 100% of labels, so the effect size is not resolved. The explanation analysis covers 64 volumes. This suffices to show that the deletion metric fails to discriminate between methods. Resolving the weak correlation between uncertainty and faithfulness would require a larger sample. The deletion baseline replaces voxels with the dataset mean; a blurred or inpainted baseline would remain nearer the data distribution and may change the conclusion of Section III-C. Inputs are 64^3 patches. A model pretrained on complete scans may therefore be used outside the regime it was designed for. The deep ensemble has four members, which is at the low end for stable uncertainty estimates, and its members share a pretrained initialisation, so their diversity arises from data ordering, subsampling and dropout only.

V. CONCLUSION

Self-supervised CT pretraining gave a positive mean AUC gain at every label budget over three paired seeds, with the gain smaller than the between-seed spread at two of the three budgets. The standard deletion protocol ranked all four attribution methods below a random-order baseline. We attribute this to distribution shift introduced by uniform voxel masking. Faithfulness metrics developed for 2D natural images require manifold-preserving masking before they can rank attribution methods on volumetric medical data. Reporting the random control is necessary, since without it the same measurements would support a ranking of the four methods.

REPRODUCIBILITY

Code, result files and all figures are available at <https://github.com/Joana-Mansa/ct-fm-lrp-uncertainty>. The full ablation runs in approximately 14 minutes on one A100. On Volta hardware, `torch` must be pinned to 2.5.1+cu121; releases from 2.14 ship an architecture list beginning at `sm_75` and every CUDA call fails.

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